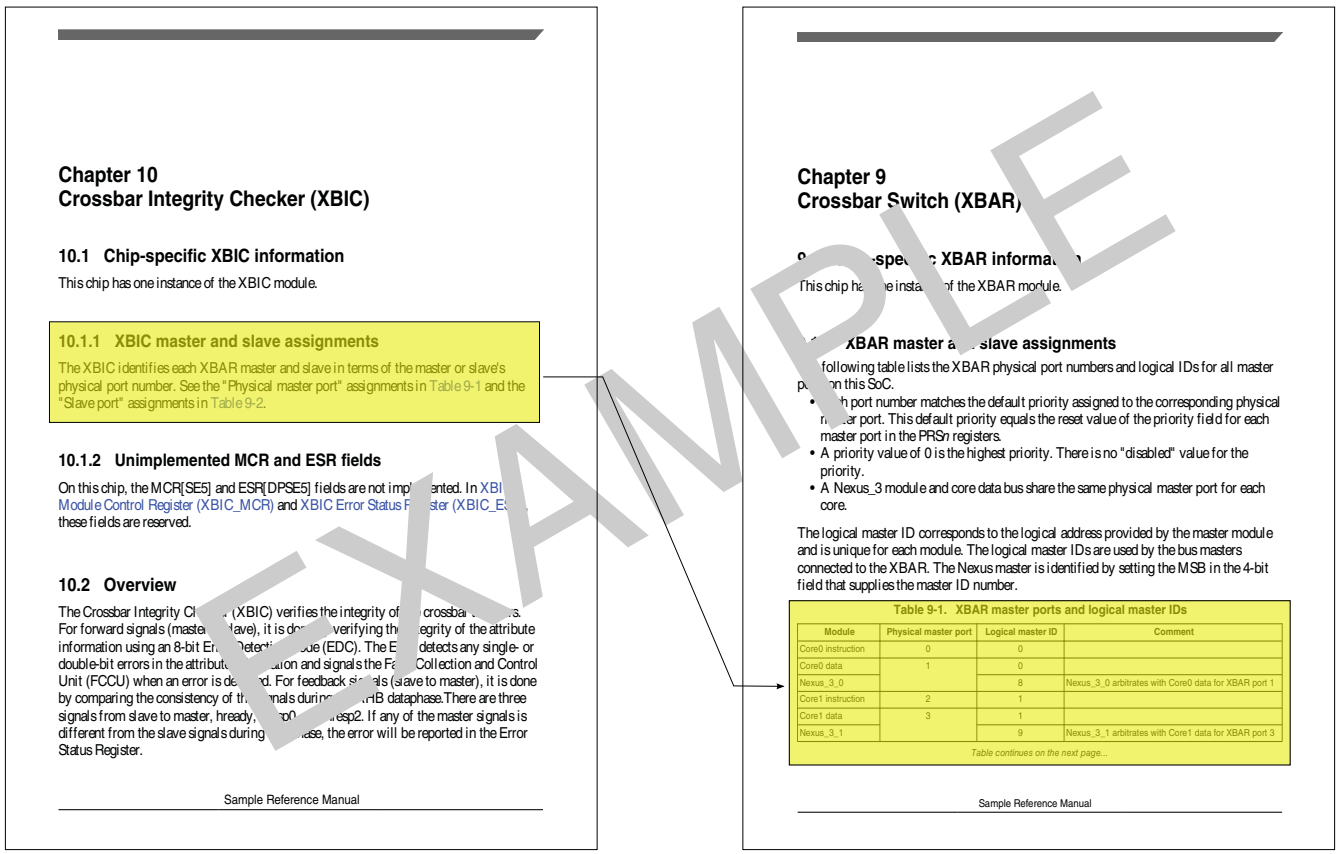


Chapter 10
Crossbar Integrity Checker (XBIC)

10.1 Chip-specific XBIC information

This chip has one instance of the XBIC module.

10.1.1 XBIC master and slave assignments

The XBIC identifies each XBAR master and slave in terms of the master or slave's physical port number. See the "Physical master port" assignments in Table 9-1 and the "Slave port" assignments in Table 9-2.

10.1.2 Unimplemented MCR and ESR fields

On this chip, the MCR[SE5] and ESR[DPSE5] fields are not implemented. In XBIC Module Control Register (XBIC_MCR) and XBIC Error Status Register (XBIC_ES... these fields are reserved.

10.2 Overview

The Crossbar Integrity Ch... (XBIC) verifies the integrity of ... crossbar ... For forward signals (maste... lave), it is do... verifying th... egrity of the attribute information using an 8-bit Er... Detecti... ue (EDC). The E... detects any single- or double-bit errors in the attribut... ation and signals the Fa... Collection and Control Unit (FCCU) when an error is de... d. For feedback si...ls (slave to master), it is done by comparing the consistency of th...nals during ...hB dataphase.There are three signals from slave to master, hready, ...sp0, ...esp2. If any of the master signals is different from the slave signals during ...ase, the error will be reported in the Error Status Register.

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Chapter 9
Crossbar Switch (XBAR)

9 ...-spe... : XBAR informa...

This chip ha... e insta... of the XBAR module.

... XBAR master a... slave assignments

The following table lists the XBAR physical port numbers and logical IDs for all master p... on this SoC.

- ...h port number matches the default priority assigned to the corresponding physical ...er port. This default priority equals the reset value of the priority field for each master port in the PRSn registers.
- A priority value of 0 is the highest priority. There is no "disabled" value for the priority.
- A Nexus_3 module and core data bus share the same physical master port for each core.

The logical master ID corresponds to the logical address provided by the master module and is unique for each module. The logical master IDs are used by the bus masters connected to the XBAR. The Nexus master is identified by setting the MSB in the 4-bit field that supplies the master ID number.

Table 9-1. XBAR master ports and logical master IDs

Module	Physical master port	Logical master ID	Comment
Core0 instruction	0	0	
Core0 data	1	0	
Nexus_3_0		8	Nexus_3_0 arbitrates with Core0 data for XBAR port 1
Core1 instruction	2	1	
Core1 data	3	1	
Nexus_3_1		9	Nexus_3_1 arbitrates with Core1 data for XBAR port 3

Table continues on the next page...

Sample Reference Manual

EXAMPLE

Figure 1-3. Example: chip-specific information that refers to a different chapter

1.4 Register descriptions

Module chapters present register information in:

- Memory maps containing:
  - An offset from the module's base address
  - The name and acronym/abbreviation of each register
  - The width of each register (in bits)
  - Each register's reset value
  - The page number on which each register is described
- Register figures
- Field-description tables
- Associated text

The register figures show the field structure using the conventions in the following figure.

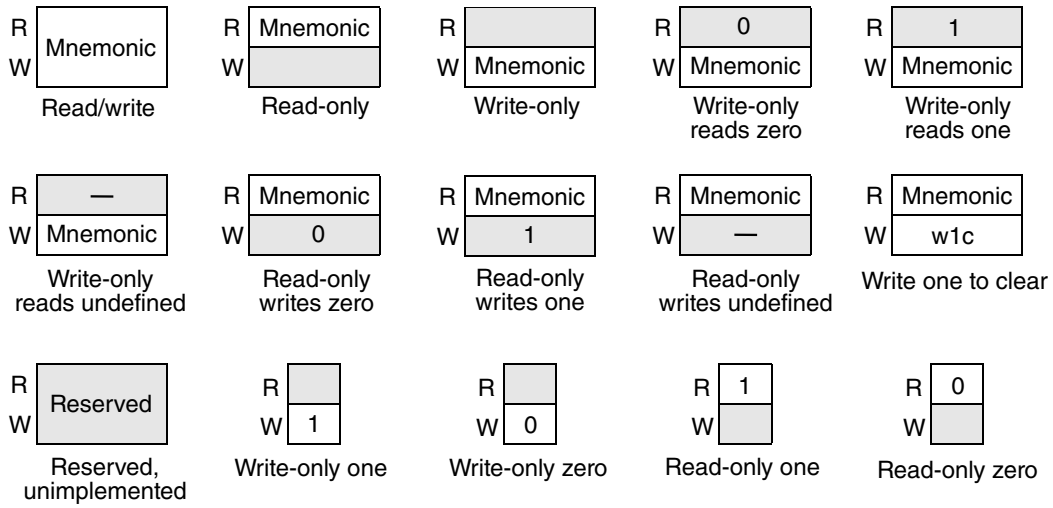


Figure 1-4. Register figure conventions

1.5 Conventions

1.5.1 Notes, Cautions, and Warnings

Note, Caution, and Warning notices appear throughout this manual. Each notice type alerts readers to a specific kind of information.

NOTE

Notes convey information that may be tangential to a topic or that may not apply to all readers.

CAUTION

Caution notices call out information that readers should know before taking further action. Cautions frequently point to trouble spots that may damage the chip or board.

WARNING

Warning notices inform readers about actions that could result in unwanted consequences, especially those that may cause bodily injury.

1.5.2 Numbering systems

The following suffixes identify different numbering systems:

This suffix	Identifies a
b	Binary number. For example, the binary equivalent of the number 5 is written 101b. In some cases, binary numbers are shown with the prefix *0b*.
d	Decimal number. Decimal numbers are followed by this suffix only when the possibility of confusion exists. In general, decimal numbers are shown without a suffix.
h	Hexadecimal number. For example, the hexadecimal equivalent of the number 60 is written 3Ch. In some cases, hexadecimal numbers are shown with the prefix *0x*.

1.5.3 Typographic notation

The following typographic notation is used throughout this document:

Example	Description
placeholder, x	Items in italics are placeholders for information that you provide. Italicized text is also used for the titles of publications and for emphasis. Plain lowercase letters are also used as placeholders for single letters and numbers.
`code`	Fixed-width type indicates text that must be typed exactly as shown. It is used for instruction mnemonics, directives, symbols, subcommands, parameters, and operators. Fixed-width type is also used for example code. Instruction mnemonics and directives in text and tables are shown in all caps; for example, BSR.
SR[SCM]	A mnemonic in brackets represents a named field in a register. This example refers to the Scaling Mode (SCM) field in the Status Register (SR).
REVNO[6:4], XAD[7:0]	Numbers in brackets and separated by a colon represent either: • A subset of a register's named field For example, REVNO[6:4] refers to bits 6–4 that are part of the COREREV field that occupies bits 6–0 of the REVNO register. • A continuous range of individual signals of a bus For example, XAD[7:0] refers to signals 7–0 of the XAD bus.

1.5.4 Special terms

The following terms have special meanings:

Term	Meaning
asserted	Refers to the state of a signal as follows: • An active-high signal is asserted when high (1). • An active-low signal is asserted when low (0).
deasserted	Refers to the state of a signal as follows: • An active-high signal is deasserted when low (0). • An active-low signal is deasserted when high (1). In some cases, deasserted signals are described as *negated*.

Table continues on the next page...

Term	Meaning
reserved	Refers to a memory space, register, field, or programming setting. Device operation is not guaranteed when reserved locations are written to any value. • Do not modify the default value of a reserved programming setting, such as the reset value of a reserved register field. • Consider undefined locations in memory to be reserved.
w1c	Write 1 to clear: Refers to a register bitfield that must be written as 1 to be "cleared."